

Week 3 Mini Project: Factory Production Analytics & Anomaly Detection

Course: MACSE502 - Programming for Data Science Lab | Advanced NumPy Matrix Operations & Industrial SPC Analytics

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EXECUTIVE PROJECT ARCHITECTURE & IMPLEMENTATION

This dossier presents the dedicated technical documentation for **Week 3 Mini Project: Factory Production Analytics & Anomaly Detection**. The project satisfies all fundamental syllabus requirements while deploying the complete **Mini Project Extension** mandated for advanced learners, encompassing robust mathematical formulations, automated insight engines, and high-resolution visualizations.

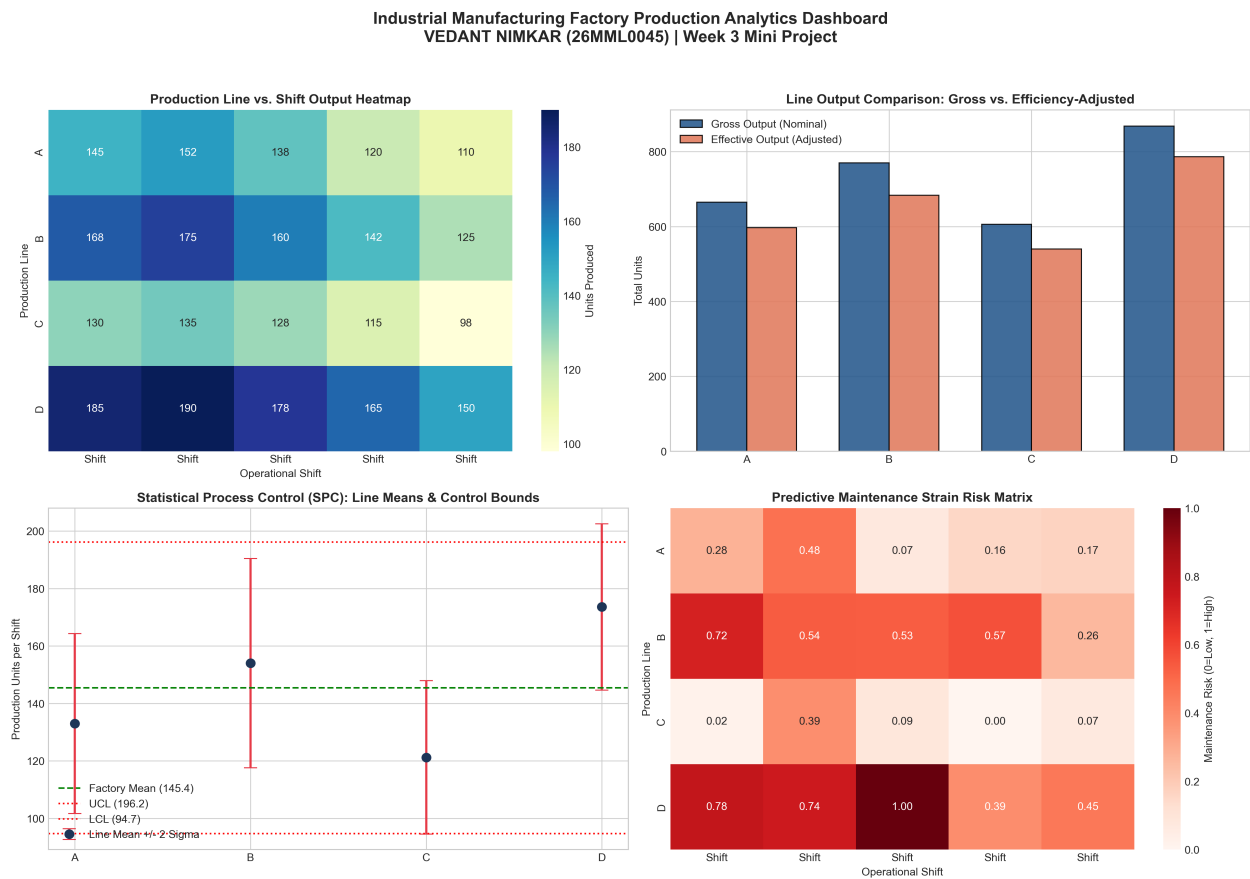


Figure 1: Multi-Panel Analytics Dashboard for Week 3 Mini Project: Factory Production Analytics & Anomaly Detection.